

Kalshi KXBTC15M “Done-Deal” Bot — Backtest Report (CORRECTED + Updated)

Chronological compounding fixed · Entry window extended to 1.75–5.0m · May 17 – Aug 15 2026
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0. Corrections & Updates in This Version

A. Compounding bug (fixed): Original trade logs compounded cash while iterating August→May. Early trades were sized off capital not yet earned. Total PnL almost unchanged; **max drawdown was overstated** (~35–50% too high at elevated risk). All path-dependent numbers below use May→August order.

B. Entry window update (applied to live bot): After testing alternate time bands, the live bot was changed from **1.75–4.5** to **1.75–5.0 minutes left**. The 4.5–5.0 slice adds ~296 trades at equal or better quality than the core. Constants `ENTRY_TIME_LEFT_MIN=1.75` and `ENTRY_TIME_LEFT_MAX=5.0` now drive the entry gate (commit message in artifacts).

1. Strategy Overview (Layman’s Terms)

Kalshi runs continuous 15-minute binary markets: “Will Bitcoin go up in the next 15 minutes?” Contracts trade 1¢–99¢. The bot only enters late in the window when (1) the ask is already 93–95¢ and (2) live BTC has moved far enough in that direction with aligned momentum and no large opposing wicks. It holds to settlement. In-sample win rate ≈ 97.8–98.0%. It is a selective “almost-certain outcome” collector, not a scalper or pure directional BTC trader.

2. Exact Parameters (from live bot.py — post-update)

- **Time left:** `ENTRY_TIME_LEFT_MIN 1.75 ≤ t ≤ 5.0 ENTRY_TIME_LEFT_MAX` (was 4.5)
- **Price:** ask in [93¢, 95¢] · prefer higher ask if both sides qualify
- **BTC filters (Binance 1m):** distance from window-open ≥ 0.10% (>3.5 min left) or 0.06%; momentum aligned; no large opposing wick
- **Sizing:** risk % of cash · optional \$500 notional cap · $qty = \text{floor}(\text{notional} \times 100 / \text{fill_cents})$
- **Costs:** fill = signal ask + 1¢ adverse slip · taker fee $0.07 \cdot p \cdot (1-p)$ ceil to next cent
- **Stops:** 20% (wider for ≥97¢); 3 consecutive 1m breaches + BTC reversal ≥50%; disabled late in window

3. Data

8,508 finalized KXBTC15M markets · 135,985 Kalshi 1m candles · ~131k Binance BTCUSDT 1m bars · May 17 – Aug 15 2026. One entry max per market (first qualifying bar in the time window).

4. Corrected Results — Original Window (1.75–4.5 min)

All figures: 1¢ slip + real fees + stops · chronological compounding from \$5,000.

With \$500 hard position cap

Risk	Trades	Win%	Total PnL	Final	Max DD (old)	Max DD (corr.)	PF
1%	509	97.8%	+\$669	\$5,669	\$98	\$89	2.16
2%	509	97.8%	+\$1,432	\$6,432	\$224	\$184	2.18
5%	509	97.8%	+\$4,314	\$9,314	\$795	\$518	2.21
7%	509	97.8%	+\$5,878*	\$10,878	—	\$738	2.18
10%	509	97.8%	+\$6,384	\$11,384	\$885	\$885	2.16

*7% with cap binds hard; pure-% figures below are preferred for uncapped scaling.

Pure % risk — no \$500 cap (use these drawdowns for risk choice)

Risk	Trades	Win%	Total PnL	Final Cash	Max DD (old)	Max DD (corrected)	PF
1%	509	97.8%	+\$669	\$5,669	\$98	\$89	2.16
2%	509	97.8%	+\$1,432	\$6,432	\$224	\$184	2.18
5%	509	97.8%	+\$4,314	\$9,314	\$795	\$518	2.21

Risk	Trades	Win%	Total PnL	Final Cash	Max DD (old)	Max DD (corrected)	PF
7%	509	97.8%	+\$6,848	\$11,848	\$1,398	\$825	2.23
10%	509	97.8%	+\$11,816	\$16,816	\$2,794	\$1,425	2.26

5. Entry-Window Expansion Tests (Key Finding)

Same filters/costs/stops; only the allowed time_left band changes. All runs chronological, pure % risk, \$5k start, 5% risk row used for comparison.

Window	Trades	Win%	Stops	Total PnL @5%	Max DD	PF
1.75–4.5 (original)	509	97.8%	0	+\$4,314	\$518	~2.2
1.0–6.0 (wide)	1,169	97.3%	4	+\$11,588	\$1,123	1.92
1.5–5.0	783	98.0%	1	+\$9,276	\$606	2.71
1.75–5.0	783	98.0%	1	+\$9,276	\$606	2.71
2.0–5.0	783	98.0%	1	+\$9,276	\$606	2.71

1.5–5.0, 1.75–5.0, and 2.0–5.0 produce the **same 783 trades**: with 1-min bars and “first qualifying bar in window,” almost no extra markets appear between 1.5 and 2.0. The entire 509→783 gain is from raising the upper bound to 5.0.

Unit-size edge by time_left (isolates quality, not compounding)

Segment	Trades	Win rate	Avg PnL (1 lot)	Notes
Core 1.75–4.5	487	97.9%	+\$0.018	Original quality
Extension 4.5–5.0	296	98.0%	+\$0.021	Equal or better than core
1.5–2.0 (early)	thin	97.1%	+\$0.010	Weaker avg PnL
5–6 (from wide test)	400	diluted overall	—	Adds volume, hurts WR/DD

Decision applied to live bot

Keep lower bound 1.75 (or 2.0); extend upper bound 4.5 → 5.0. Doubles opportunity and roughly doubles \$ PnL at 5% risk vs original, with only a modest DD increase (\$518→\$606). Avoid 1.0 and 6.0 extremes (lower quality / higher path risk).

Risk sweep — recommended window 1.75–5.0 (same as 2.0–5.0 in sample)

Risk	Trades	Win%	Stops	Total PnL	Final Cash	Max DD	PF
1%	783	98.0%	1	+\$1,177	\$6,177	\$99	2.45
2%	783	98.0%	1	+\$2,635	\$7,635	\$204	2.51
5%	783	98.0%	1	+\$9,276	\$14,276	\$606	2.71
7%	783	98.0%	1	+\$16,469	\$21,469	\$1,195	2.85
10%	783	98.0%	1	+\$34,012	\$39,012	\$2,817	3.09

6. Monthly PnL — Original 1.75–4.5 @ 5% pure risk (corrected path)

Month	Trades	Win rate	PnL
2026-05	65	96.9%	+\$229
2026-06	241	97.5%	+\$1,490
2026-07	146	97.9%	+\$1,387
2026-08 (partial)	57	100.0%	+\$1,208
Total	509	97.8%	+\$4,314

7. Day-of-Week — Original window @ 5% (corrected path)

Day	Trades	Win rate	Avg PnL	Total PnL
Thursday	71	95.8%	+\$2.07	+\$147
Tuesday	82	96.3%	+\$2.82	+\$231
Sunday	44	97.7%	+\$6.74	+\$296
Friday	85	97.6%	+\$9.67	+\$822
Monday	98	99.0%	+\$11.26	+\$1,104
Wednesday	91	98.9%	+\$12.81	+\$1,166
Saturday	38	100.0%	+\$14.43	+\$548

Win-rate ranking is the stable signal: Thursday & Tuesday weakest; Saturday strongest in-sample.

8. Capital for ~\$500 / Month (Original Window Edge, Corrected)

Risk	Avg monthly @ \$5k	Capital for ~\$500/mo
1%	~\$223	~\$11,200
2%	~\$477	~\$5,240
5%	~\$1,438	~\$1,740
7%	~\$2,283	~\$1,100
10%	~\$3,939	~\$635

With the **1.75–5.0** window, monthly \$ at the same risk is substantially higher (more trades). At 5% risk the backtest produced ~\$3,000+/month equivalent on \$5k over the sample — scale linearly, but treat as optimistic.

9. Projected Monthly Profit by Balance (Original 1.75–4.5 Edge)

Starting Balance	2% risk	5% risk	7% risk
\$1,000	~\$95	~\$290	~\$460
\$2,000	~\$190	~\$575	~\$915
\$5,000	~\$475	~\$1,440	~\$2,280
\$10,000	~\$950	~\$2,880	~\$4,560
\$15,000	~\$1,425	~\$4,320	~\$6,840

10. Live Bot Code Change

File: bot.py

Added constants:

```
ENTRY_TIME_LEFT_MIN = 1.75  
ENTRY_TIME_LEFT_MAX = 5.0 # was hard-coded upper 4.5
```

Entry gate:

```
if ENTRY_TIME_LEFT_MIN <= time_left <= ENTRY_TIME_LEFT_MAX and (y_qualifies or n_qualifies):
```

Startup log now prints the active band. Commit (artifacts repo): extend window 1.75–4.5 → 1.75–5.0 with backtest rationale in message.

11. Caveats (Include in Any Downstream Analysis)

- 1. Fill assumption is optimistic vs live-adjacent data.** Backtest fills 100% of signals at signal_ask+1¢. Live bot pessimistic shadow has shown ~69–83% fill rates with slippage both ways.
- 2. Possible data-snooping on the 93–95¢ band** (and to a lesser degree the new 5.0 upper bound). Both were informed by observations overlapping this sample. Not independent out-of-sample. Walk-forward would strengthen claims.
- 3. Regime / sample risk.** June was unusually active. Quiet BTC periods reduce opportunity. Past edge ≠ future edge.
- 4. Window test is still in-sample.** Choosing 5.0 because it looked good on this window is a mild form of selection; monitor live fill quality and win rate after deploy.

12. Concise Summary for Claude / Other Analysts

Done-deal bot on Kalshi KXBTC15M: enter only when 1.75–5.0 min remain, ask is 93–95¢, and Binance BTC confirms distance + momentum + no large opposing wick. Hold to settlement (stops almost never fire on these setups).

Corrected backtest (May 17–Aug 15 2026, chronological sizing): original 1.75–4.5 window → 509 trades, 97.8% win, +\$4.3k at 5% risk on \$5k, max DD ~\$518. Extending upper bound to 5.0 → 783 trades, 98.0% win, +\$9.3k, max DD ~\$606. The 4.5–5.0 minute slice is at least as high-quality as the core. Live bot updated accordingly.

Do not use the first report's max-drawdown column for risk sizing; use corrected (lower) DDs. Treat fill rates and parameter choices as optimistic relative to true out-of-sample.

CORRECTED + window-update report · Use for risk decisions and for sharing with other models · Artifacts: bot.py (updated), signals caches, trade CSVs under /home/workdir/artifacts/